

Modular Battery System For Solar Applications

The untransferred energy in a solar battery system can result into premature aging of the solar cells. A Maximum Power Point (MPP) converter may provide a partial solution to the problem, but it needs to be redesigned to survive a battery failure. Read on to know more

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If you take a look in most hardware stores, you will find a small solar panel that is designed to keep seldom-used batteries charged. Whether the battery is a small, rechargeable NiCd for a radio or a deep-cycle marine battery for a boat, the basic principal, and circuit, are the same. Essentially, you'll find an array of solar cells connected to the battery through a rectifier diode. The diode prevents the battery from reverse biasing the solar cell when there is no sun, and the array of solar cells produces a slightly higher voltage than the charged voltage of the battery—so it can forward bias the diode and charge the battery. This system neatly solves the problem of maintaining a charge in battery-based systems, without the need for a line connection.

While this system is certainly simple, it does suffer from a few problems in efficiency. The main problem is that the optimum loading for the solar cell is dependant upon how much sun light it receives. Just strapping on a battery doesn't guarantee that the cell will be seeing its optimum load. In fact, the voltage across the battery will vary as it charges, so there is probably energy that is not being transferred to the battery in full light conditions. And, whatever doesn't get used to charge the battery remains in the solar cell as heat, prematurely aging the solar cells.

In addition, just getting the energy to the battery does not guarantee that the battery will have the proper ca-



capacity to store the energy. Once fully charged, continuing to charge the battery just dumps heat into the battery instead of the solar cell, resulting in the battery prematurely aging.

To achieve a more efficient solution, we need something that can present the optimum load to the solar cells, while charging the battery efficiently. It turns out that there is a partial solution to this problem; it is called a Maximum Power Point, or MPP, converter. The MPP converter looks like a switching regulator with some rudimentary intelligence driving the current set point. It boosts or bucks the solar cell voltage up/down to the battery voltage, while regulating the load it presents to the solar cells, such that the maximum amount of power is transferred. It does this by periodically varying the current

that the switching regulator draws from the solar cells in a process called dithering. In dithering, the regulator increases the load current, then decreases it. If the higher current results in more power from the cell, then the higher current draw is retained. If a lower current gives more power, then the lower current is retained. If neither a higher nor lower load results in higher power levels, then the original loading is retained.

The overall result is that the MPP converter draws the maximum power possible from the solar cells, and if the amount of sun light changes, then the periodic dithering will adjust the current until it is once again pulling the maximum power. The MPP doesn't worry too much about heat loss in the battery as, frankly, its limited intelligence is optimised for the solar array. Batteries are significantly cheaper than solar cells, and will have to be periodically replaced anyway—so any extra energy is still dumped into the batteries (though this amount is much smaller with an MPP).

So, we have a solar-cell array that is driving a MPP converter and a battery. If the system is to be run off of the battery voltage, then another switching regulator is likely needed to convert the variable battery voltage into a steady voltage for the load. (See Fig. 1.)

The overall efficiency of the system